Verification 03_H_Simulator_FVM_1D_Compressible_Homogeneous

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional slightly compressible single-phase flow in a homogeneous porous medium.

2 Governing Equation

For one-dimensional slightly compressible single-phase flow in a homogeneous porous medium, conservation of mass combined with Darcy's law gives

$$\frac{\partial(\phi\rho)}{\partial t} = \frac{\partial}{\partial x}\left(\frac{\rho k}{\mu}\frac{\partial P}{\partial x}\right), \tag{1}$$

where

- P is pressure [Pa],
- ρ is fluid density [kg/m^3],
- ϕ is porosity [-],
- k is permeability [m^2],
- μ is fluid viscosity [Pa·s].

For a slightly compressible fluid,

$$c_f = \frac{1}{\rho}\frac{\partial\rho}{\partial P}, \tag{2}$$

and assuming constant porosity and compressibility, the density is given by

$$\rho = \rho_0 e^{c_f(P-P_0)}, \tag{3}$$

where ρ_0 is the reference density at pressure P_0. For small pressure changes, the governing equation may be linearized to obtain the pressure diffusivity equation,

$$\frac{\partial P}{\partial t} = \alpha\frac{\partial^2 P}{\partial x^2}, \tag{4}$$

where

$$\alpha = \frac{k}{\phi\mu c_f} \tag{5}$$

is the hydraulic diffusivity. The domain is defined on

$$0 \le x \le L_x, \tag{6}$$

with fixed-pressure boundary conditions

$$P(0,t) = P_L, \qquad P(L_x,t) = P_R, \tag{7}$$

and the initial condition

$$P(x,0) = P_i. \tag{8}$$

3 Analytical Solution

To validate the numerical simulator, a semi-infinite analytical solution is used. This is justified because, for the chosen simulation time and domain length, the pressure diffusion front does not reach the right boundary, making boundary effects negligible. The governing equation is the one-dimensional diffusion equation

$$\frac{\partial P}{\partial t} = \alpha \frac{\partial^2 P}{\partial x^2}, \tag{9}$$

with hydraulic diffusivity

$$\alpha = \frac{k}{\phi \mu c_f}. \tag{10}$$

The domain is considered semi-infinite

$$x \geq 0, \tag{11}$$

with initial condition

$$P(x,0) = P_i, \tag{12}$$

and a constant boundary pressure imposed at the origin

$$P(0,t) = P_L. \tag{13}$$

3.1 Semi-infinite solution

The analytical solution is given by the error function formulation

$$P(x,t) = P_i + (P_L - P_i)\,\mathrm{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right). \tag{14}$$

3.2 Validity of the approximation

The semi-infinite assumption is valid when the characteristic diffusion length

$$\delta(t) \sim \sqrt{\alpha t} \tag{15}$$

is significantly smaller than the domain length L_x, i.e.

$$\sqrt{\alpha t} \ll L_x. \tag{16}$$

Under the present simulation conditions, this criterion is satisfied, and comparison against the finite-domain solution confirms that both formulations produce indistinguishable results within numerical accuracy.

4 Simulation Parameters

The simulation parameters are given in Table 1.

Parameter	Symbol	Value
Number of cells	N_x	10000
Domain length	L_x	10 m
Permeability	k	1×10^{-12} m^2
Viscosity	μ	0.1 Pa·s
Left pressure	P_L	10×10^6 Pa
Right pressure	P_R	1×10^5 Pa
Compressibility	c_f	5×10^{-8} Pa^{-1}
Reference density	ρ_0	1000 kg/m^3
Reference pressure	P_0	1×10^5 Pa

Table 1: Simulation parameters.

5 Results

5.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

6 Error Analysis

The maximum pressure relative error is defined as

$$E_P = \max_i \left| \frac{P_{\text{num},i} - P_{\text{ana},i}}{P_L - P_R} \right|. \tag{17}$$

For the present simulation in MATLAB

$$E_P = 0.0427. \tag{18}$$

In Python, the result is

$$E_P = 0.0427. \tag{19}$$

7 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution and the numerical Darcy flux reproduces the analytical constant flux solution. The small residual errors arise from floating-point arithmetic and are negligible.

8 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional slightly compressible flow in a homogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

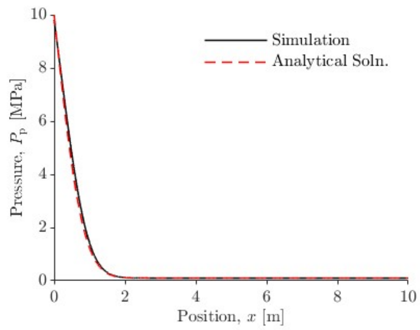

Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

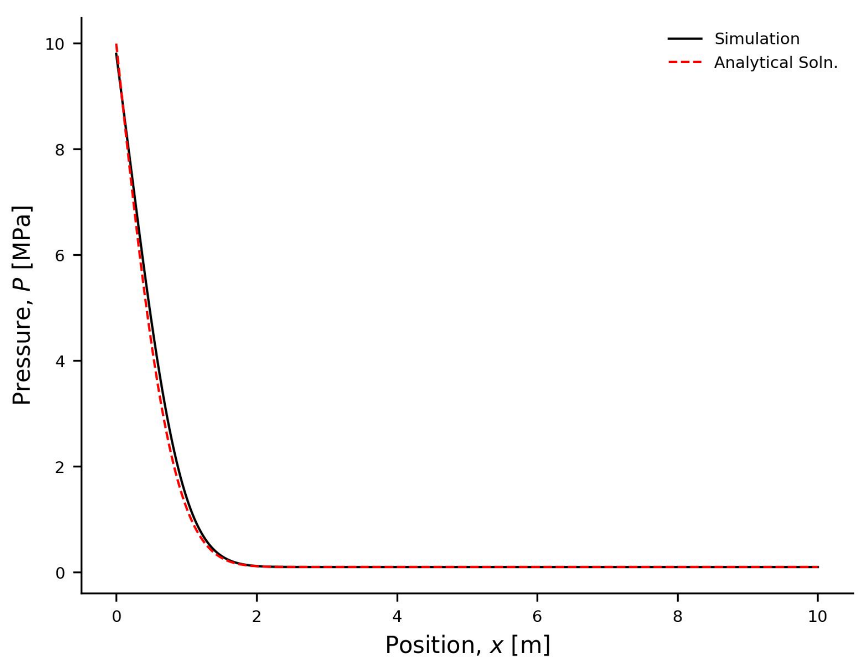

Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.